

State of *CryptAI*

A nine-specialist audit of the repo, the deployment, the cost model, and the quant evidence — every claim file:line-verified on branch `fix/m1-foundation-bugs` @ 8cb897e, 2026-08-24.

Doc	Method	Verdict
01 of 06 — Audit	9 parallel audit agents · 325 tool calls	Product built · operations broken · fixable in days

The product is built. The operations broke it. Both are fixable.

TESTS GREEN 912 687 backend · 99 frontend · 126 Rust — verify.sh passes in 38s	BACKEND SUSPENDED Render bandwidth overage 5.63/5 GB, since 2026-08-10	FRONTEND LIVE cryptai-app.vercel.app · Vercel Hobby	MONTHLY SPEND \$0 every service on a free tier — blocked, not billed	SINCE LAST AUDIT 43 commits — the session pipeline went from unwired to wired
WHAT CHANGED SINCE THE AUG-2 AUDIT — THE GOOD NEWS The Aug-2 verdict was "three good products that never met." That gap is closed in code: the session pipeline (start → per-user synthesis → proposal → approve → re-validate at live price → execute → per-position monitor) is genuinely wired into the one production entrypoint, with per-session LLM billing feeding a fail-closed cap. The Rust engine ships inside the production Docker image. Confirmed bugs #1–4 from the old audit (quota bypass, admin fail-open, unfed risk tracker, Delta crash) are fixed with traced call paths.			WHY USERS SEE A DEAD PRODUCT ANYWAY • None of the 43 commits is deployed — Render suspended the workspace before they could ship. • The suspension's root cause is only half-fixed: the API's market feed was gated, but the daemon's DataAgent still opens ~2.2 GB/day of streams at startup (slide 4). • Supabase has been idle 14 days — past its 7-day free-tier pause threshold; it likely needs a dashboard restore too. • A newly found bug means the signal plane cannot simply be switched on (slide 5).	

Subsystem truth table — construction-path verified, not test-verified

SUBSYSTEM	STATUS	EVIDENCE-BACKED REALITY	CAVEAT THAT MATTERS
Session pipeline	WIRED	Pool constructed in the daemon, daemon embedded in the API process; a metered session runs real analysis and creates proposals; clock pauses on propose	—
Per-user synthesis	WIRED	LLM selects the candidate fitting <i>this</i> trader + writes the thesis; never invents prices; forecast-language lint; deterministic fallback	Candidate <i>generation</i> still shared per symbol — selection is per-user
Approve → execute	WIRED	Re-validates expiry/invalidation/price-drift at live price, books through the user's risk gate; lost-reply recovery; 30s sweeper repairs orphans	—
Position monitors	WIRED	Supervisor unconditional; 1 unmetered monitor per position: time stop, profit give-back, adverse-pulse exit, reduce-only close	Do not survive a daemon restart — paper engines never rehydrate; Render restarts are routine
Session LLM billing	WIRED	Usage normalized + booked even on failure paths, integer micro-USD, cap ends the session fail-closed	All-:free rotation prices at \$0 — the time quota is the only binding meter today
Risk limits	FIXED	PortfolioStateTracker fed by both production booking paths — free-tier caps now evaluate a real portfolio	Live (non-paper) close hook still missing — errs over-strict, the safe direction
Rust signal engine	DARK	Built into the production image, supervised child-process wiring complete, 126 tests	Never enabled: <code>RUN_SIGNAL_ENGINE_IN_API</code> unset — and see the Pulse bug (slide 5)
Hot-window surface	ABSENT	No global pulse, no tradability history: <code>pulse_snapshots</code> has a table, indexes, a parity test — and zero writers	The founder's "hottest hour" has no data source until the writer ships
Tier/quota model	GAP	Every tier gets the same pooled 30 min/day; session count unlimited; channels not tier-gated; nothing blocks a new session while a trade is open	The 2x30-min slot model + focus lock is R2 work (doc 03/06)
Legacy single-bot stack	LEGACY	<code>execution_agent</code> / <code>order_tracker</code> / old monitor: tested but reachable only from <code>src/main.py</code> , which production never runs	Marked legacy; candidates for a later attic pass

The bandwidth post-mortem: streams that served nobody

Render free allows 5 GB/month of bandwidth — ~167 MB/day. On Aug 10 the workspace hit 5.63 GB, of which **5.61 GB was service-initiated** (17 MB actually served users). Two always-on Binance consumers did it — kept alive 24/7 by our own keepalive cron.

CONSUMER	COST	STATE ON HEAD	FIX
Daemon DataAgent — depth20@100ms + markprice@1s + ticker + 5 kline TFs per symbol, subscribed unconditionally at startup	~2.2 GB/day	STILL IN CODE — only the LLM analysis was demand-gated, never these streams. Resume today → re-suspended in ~2–3 days	Gate to scan demand + REST snapshots (R0, before resume)
API base market feed — btcusdt ticker + depth5@100ms + kline_1m, formerly subscribed at boot streaming to nobody	~216 MB/day	FIXED — refcounted to live dashboard viewers, 30s linger (commit 9edada0); idle backend subscribes nothing	Deployed on resume
Engine bookTicker stream — pushed on every best-bid/ask change	~200 MB/day	FIXED — removed from the stream URL with a regression test; replaced by one batched REST poll per 5s	Deployed when engine ships
Engine kline WS — 12 streams at 3 symbols × 4 TFs	~245 MB/day	OVER BUDGET IF ENABLED — 7.4 GB/mo alone	REST-poll closed bars (R1) → ~2 MB/day
Idle-socket churn — zero-stream socket trips its own 90s staleness check, reconnects forever	~6 MB/day	NEW FINDING — ~960 pointless TLS handshakes/day + log noise	Skip staleness when no subscriptions (R0)

IF RESUMED AS-IS

~2.4 GB/day

14× budget — suspension #2 within days

AFTER R0 + R1 FIXES

~1.5 GB/mo

viewer-gated feed + REST polling — 3.5 GB monthly headroom

THE LAW GOING FORWARD

Name your consumer

no stream or loop without a live session, viewer, or open position attached

The defect list that matters now

#	DEFECT	SEVERITY	CONSEQUENCE IF SHIPPED	WHEN
1	Daemon DataAgent always-on streams	BLOCKER	Re-suspension within days of resuming — product dies again	R0 · pre-resume
2	Pulse dataclass vs <code>.get()</code> — new find, reproduced	HIGH	Enabling the signal plane fails every metered cycle while the user's clock drains; tests pass only because they fake dicts	R1 · pre-enable
3	Monitors + paper desks lost on restart — engines never rehydrate	HIGH	"Your agent never sleeps" is false after any deploy/restart — the product's core promise breaks silently	R2
4	Symbol-universe mismatch — analysis runs BTC/ETH/XAUT, sessions run BTC/ETH/SOL	MED	~1/3 of all LLM spend produces setups no session can use; SOL sessions scan a symbol with no candidates. One-line env fix	R1 · env
5	Quota model gaps — same 30 min/day for every tier, unlimited session count, no focus lock, channels not tier-gated	MED	The tier ladder is unenforceable; free users can start sessions against an open trade and burn clock on undeliverable setups	R2
6	NEXT_PUBLIC_API_TOKEN — admin-capable static token labeled "read-only", plumbed as a frontend build arg	MED	One operator mistake ships an admin token in the public JS bundle	R3
7	Fail-open anonymous mode — no SUPABASE_URL + no API_AUTH_TOKEN ⇒ everyone sees all tenants' trades, and is admin	MED	Tenancy rests entirely on env vars being present	R3
8	WS JWT in <code>?token=</code> + public <code>/metrics</code>	LOW	JWTs land in proxy access logs; internals exposed	R3
9	188 quarantined tests (~21% of the suite)	LOW	Green pytest overstates coverage; precedent shows some quarantined tests encode real regressions	ongoing

Fixed & verified since Aug 2 (for the record): session-quota clamp · admin fail-open for end-users · risk tracker fed by real bookings · Delta OrderStatus crash · session LLM billing. The uncommitted working-tree diff (analysis demand-gate completion) is **correct and worth keeping** — commit it after fixing its stale docstring and aligning `CYCLE_INTERVAL=180`.

Dead weight is out — reference-checked, attic'd, gate still green

ONE-OFF SCRIPTS RETIRED

44

incl. nuclear_clear_positions.py & friends — **out of the production image**; scripts/admin.py is the guarded path

DEAD SRC MODULES

6

zero importers from any entrypoint or test — incl. EventCoordinator

SUPERSEDED DOCS

35+

Jan-2026 doc generation + 12-file fix-summary
citation island

GATE AFTER THE MOVE

GREEN

verify.sh: typecheck ✓ 99 frontend ✓ import
smoke ✓ 687 backend ✓

ABOUT THAT "300-SECOND ALWAYS-RUNNING LOOP"

The founder remembered an unused 300s loop. Found it: `EventCoordinator` — a 300s cycle trigger + 30s monitor. The twist: **it was never constructed anywhere**. It burned zero cycles; it was pure reading noise. The loop that actually cost money was the DataAgent's stream fleet (slide 4). Both are now handled: the corpse is in `_attic/src-dead/`, the live one is R0's blocker fix. **Lesson encoded in CLAUDE.md**: in this repo, "exists and passes tests" ≠ "runs" — always trace a constructor from an entrypoint.

WHERE EVERYTHING WENT

LOCATION	CONTENTS
<code>_attic/cta-scripts/</code>	44 one-off/destructive scripts (+ .dockerignore excludes _attic)
<code>_attic/src-dead/</code>	event_coordinator, health_monitor, cycle_state, multi_context_analyzer, dual_context_llm_builder, ict_detector_CLEAN
<code>_attic/root-docs/</code> · <code>docs/</code> · <code>cta-docs/</code>	superseded doc generations + websocket test HTML + debug scripts
<code>_attic/README.md</code>	what, why, and how to restore (git mv back)

Every move was reference-checked first; pointers in README, PRD, admin.py, data_models.py, and eslint config were updated. Nothing deleted — git remembers, and the attic travels with the repo.

LLM reality: dollars are \$0 — requests are the real currency

CALL SITE (ALL → OPENROUTER :FREE ROTATION)	CADENCE	PROMPT SIZE	NOTE
Shared market analysis	per symbol per 180s cycle, demand-gated	~40k tokens	≈95% of all token throughput. Raw JSON dumps of market data — the #1 shrink target (send the engine's pulse summary instead)
Strategy creator	per symbol per cycle, when opportunities exist	~1–2k	Its purpose-built cache is constructed, tested, and has zero callers — the signature trap again; wire it or delete it
Per-session synthesis	once per session per cycle	~0.7k in / 400 out	The only session-billed call. Per-user by design, correctly uncached
Trade monitoring	15s polls	0	Fully deterministic — the "never sleeps" promise costs zero LLM

THE FREE-CAP MATH (50 USERS × FREE TIER)

One 30-min session ≈ **40–70 requests / 0.7–1.3M tokens** (\$0 on :free). OpenRouter free caps ≈ 50 req/day — or **~1,000 req/day after a one-time \$10 credit** (verify current terms). 50 users staggered across the day ≈ 3,400 req/day = dead key. **Clustered into hot windows ≈ 560 req/day = it fits.** The hot-window mechanic isn't just UX — it is the cost model.

ORDERED COST LEVERS

- **Buy the one-time \$10 OpenRouter credit** — 20× the request cap; still effectively \$0/mo
- **Shrink the 40k-token analysis prompt** to a few k (pulse summary in, raw JSON out)
- **Fix the symbol mismatch** — recovers ~1/3 of spend instantly (env one-liner)
- **Hot-window clustering** — sessions share the demand-gated analysis cycles
- **Wire the strategy cache** (built, unwired) + keep analysis cache ≥2 cycles
- **Global daily request counter, fail-closed** — nothing tracks requests today; a dead-primary rotation day multiplies the 40k prompt by every retry
- Honest floor at scale: ~\$10–20/mo of paid inference once >10–15 staggered sessions/day — a Lite-tier pricing input, not a launch cost

Quant verdict: a real filter, no directional edge, log everything

● VALIDATED — BUILD ON IT

- **Tradability = a NOW-cast filter** for move size over the next ~4–12h (IC +0.083/+0.096 at 4×1h bars; survives vol-normalization; decays to nothing by 24 bars).
- **Hour-of-day is the strongest factor** (residual IC +0.093, t +7.07) and is deterministic from the clock — the only honestly *forward* piece. London–NY overlap $\approx$ **17:30–20:30 IST**: perfect copy for Indian evening users.
- **Intraday is the only honest channel**: scalp fails validation (weights flip at 12 bars) *and* TDS economics; swing IC is insignificant at 24 bars, negative at 72.

● REFUTED / UNPROVEN — NEVER CLAIM

- **Regime→direction is dead** (walk-forward Δ IC −0.008, a coin flip). The engine enforces this on the wire; the ban must now travel **up the stack** — the LLM synthesis layer is the only source of direction and carries **zero empirical validation**. Mirror the Rust contract test in the frontend.
- **No variant shows skill after multiple-testing correction** (deflated Sharpe 0.00 everywhere); no costs/slippage/funding modeled; 2 of 6 factors unmeasured.
- Therefore: **real-money launch is indefensible**. Paper-first is not caution — it's the only position the data permits.

● DAY-ONE LOGGING (FALSIFIABILITY)

- `pulse_snapshots` writer on every publish — full factor JSON + reference price (**zero writers today**; everything downstream is blocked on this).
- Forward-return backfill (15m/1h/4h + MFE/MAE) → re-run the walk-forward at +6 weeks; if temporal IC dies, the badge comes down.
- Per-proposal log: pulse id → channel → LLM direction → user decision → paper outcome — builds the direction-layer evidence that doesn't exist.
- Hour-of-day |return| buckets per symbol → replaces the hand-set liquidity map with learned hot hours.

THE HONEST PRODUCT SENTENCE

"CryptAI ships a **validated market-conditions filter with live public calibration**, per-user trade construction, and unmetered position protection — on paper desks first, with real money gated behind six weeks of live evidence and a compliance floor." Design consequence: intraday holds (4–12h) **outlive 30-minute sessions** — the monitor isn't a feature, it's the product's spine.